

A VOCATIONAL TRAINING/INTERNSHIP REPORT

ON

“CROP PREDICTION”

A Training Report Submitted to

**CHHATTISGARH SWAMI VIVEKANANDA TECHNICAL
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BACHELOR OF ENGINEERING

In

CSE

By

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Under the Training of

AI INNOVATION

Training In-Charge

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BATCH 2016 - 2020

DECLARATION

I the undersigned solemnly declare that the Vocational Training report on **“CROP PREDICTION”** is based on my training work carried out during my vocational duration under the supervision of **KHILENDRA CHANDRAKAR, Designation** from **AI INNOVATION**.

I assert that the statements made and conclusions drawn are an outcome of the Vocational Training. I further declare that to the best of my knowledge and belief that this report does not contain any relevant work which has been submitted earlier.

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CERTIFICATE

This is to certify that the report of my vocational training on “**CROP PREDICTION**” is the work carried out by **Mr. S SHUBHAM** , studying in 7th semester in CSE branch affiliated to Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.), India under the guidance and supervision of **Trainer’s Name**.

To the best of my knowledge and belief the report

- Embodies the work of the candidate himself/herself.
- Has duly been completed.
- Fulfills the requirement of the ordinance relating to vocational training/internship w.r.t. the university curriculum.

For being referred to the examiners.

Signature

HOD (CSE)

Signature

Co-ordinator In-Charge (CSE)

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ABSTRACT

Crop prediction is a critical aspect of modern agriculture, enabling farmers to make informed decisions about crop selection, resource allocation, and cultivation practices. This project aims to develop a predictive model using Python to forecast crop yields based on various factors such as soil characteristics, weather conditions, and historical crop data.

The project employs a combination of data preprocessing, feature selection, and machine learning algorithms to build an accurate and reliable crop prediction model. Key steps include data collection from relevant agricultural databases, cleaning and normalizing the data, and applying machine learning techniques such as Random Forest, Support Vector Machines (SVM), and Artificial Neural Networks (ANN) for predictive analysis.

The model's performance will be evaluated using metrics like accuracy, precision, and recall, and fine-tuned to improve its predictive capabilities. By providing insights into the most suitable crops for specific regions and conditions, this project aims to enhance agricultural productivity and sustainability.

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LIST OF ABBREVIATION

S.No	Abbreviations	FULL FORM
1.	ML	Machine Learning
2.	AI	Artificial Intelligence
3.	SVM	Support Vector Machine
4.	RF	Random Forest
5.	ANN	Artificial Neural Network
6.	KNN	K-Nearest Neighbors
7.	LR	Logistic Regression
8.	R2	r-Squared
9.	MAE	Mean Absolute Error
10.	RMSE	Root Mean Square Error
11.	PCA	Principal Component Analysis
12.	EDA	Exploratory Data Analysis
13.	API	Application Program Interface
14.	CSV	Comma-Seperated Values

CHAPTER 1

INTRODUCTION

The global population is projected to reach 9.7 billion by 2050, putting immense pressure on our already strained food systems. Over the past few years, rapid climatic changes have been experienced. In India, this manifestation is in the form of an erratic monsoon. Various parts of the country get unpredicted amounts of rainfall and the farmers are generally unprepared. As the atmospheric conditions keep changing rapidly, farmers fall prey to the lack of knowledge that is required to estimate what kind of farm conditions, techniques and soil type is essential for growing a type of crop. Moreover, every so often it happens that farmers over utilize a particular piece of land to such an extent that it leaves the land devoid of all minerals. Hence, it is important to be able to predict and forecast the performance of the crop for all kinds of environmental conditions. Traditional agricultural practices face numerous challenges: Resource scarcity, Climate change, Increasing demand. Smart farming emerges as a revolutionary approach to address these challenges by integrating technologies like and Machine Learning etc into agricultural practices.

ML models can analyze vast datasets to identify optimal growing conditions, predict crop yields, and recommend resource allocation strategies. By analyzing real-time sensor data from the field (soil moisture, temperature, humidity), ML models can guide farmers in applying water, fertilizers, and pesticides efficiently, minimizing waste and environmental impact. ML insights can empower farmers to make informed decisions regarding planting schedules, crop selection, and resource allocation, fostering proactive and adaptable farm management strategies. By optimizing resource use and minimizing waste, smart farming contributes to a more sustainable agricultural sector, reducing the environmental footprint of food production.

1.1 Objective of the project:

The objective of the project is to develop and implement a smart farming system utilizing machine learning technologies to optimize agricultural practices and enhance the profitability and sustainability of farming operations. collecting and integrating diverse agricultural data sources, including soil characteristics, weather patterns, and crop types, to provide a comprehensive understanding of the farming environment. Developing predictive models using machine learning algorithms to forecast critical parameters such as crop yields, disease outbreaks, and weather conditions with high accuracy. Reduce resource waste by optimizing application based on specific needs of the crops and environmental conditions. By achieving these objectives, the project aims to demonstrate the transformative potential of smart farming powered by machine learning, ultimately contributing to the advancement of sustainable and resilient agricultural practices.

1.2 Problem statement:

- Traditional agricultural practices face numerous challenges:
 1. Limited accuracy in crop yield prediction
 2. Suboptimal resource utilization
 3. Reactive decision-making
- Using Smart farming ML model these issues are addressed by:
 1. Improving crop yield prediction
 2. Optimizing resource allocation
 3. Facilitating proactive and adaptable decision-making

1.3 Scope of the project:

This project focuses on using machine learning (ML) to improve crop yield prediction and resource management in smart farming. The project aims to address challenges like resource scarcity, climate change, and increasing demand for food by improving agricultural efficiency and sustainability. Additionally, the project will involve designing and deploying crop recommendation systems to suggest the most suitable crops for cultivation based on local environmental factors and market demand. The scope also includes rigorous evaluation and validation of the smart farming system's performance against historical data and real-world scenarios, as well as deployment in collaboration with farmers and stakeholders to foster adoption and ensure effective utilization. the project aims to measure the tangible benefits of the smart farming system on crop yields, resource efficiency, and economic viability, thereby contributing to the advancement of sustainable and resilient agricultural practices.

CHAPTER 2

PROJECT ANALYSIS

2.1 Existing systems:

The agriculture sector needs a huge up-gradation in order to survive the changing conditions of the Indian economy. Along with the advances in machines and technologies used in farming, useful and accurate information about different matters also plays a significant role in it. This information is being gathered by the use of remote sensors, satellite images, surveys, etc. This information along with the knowledge of subject experts and researchers should be readily available to the farmers in order to exploit its potential worth.

2.2 Issues in Existing System:

1. As the amount of such information is increasing gradually.
2. We will identify certain patterns in words slowly start understanding.
3. It also provides an insight into the troubles faced by Indian farmers and how they can be resolved using these techniques.

2.3 Literature review:

Journal paper 1:

Title: Affordable Smart Farming Using IoT and Machine Learning

Authors: Reuben Varghese, Smarita Sharma

Inference : In this paper the system leverages Internet of Things(IoT) and Machine learning to produce an affordable smart farming module. The system uses state-of-the-art methods in order to improve the accuracy of the results and automate the monitoring of crops thereby requiring minimal human intervention.

Advantages : This paper tries to reduce the losses faced by the farmers due to unpredicted weather conditions and also monitors the humidity in the soil.

Disadvantages: Computer vision features can also be added which would help us to understand the kind of soil and the type of seed using just an image. Computer vision can also be used to detect disturbances in the farm.

Journal paper 2:

Title: An IoT Based Smart Farming System Using Machine Learning

Author: A. Dahane , R. Benameur, B. Kechar , A. Benyamina

Inference : The aim of this paper is to demonstrate the effectiveness of AI techniques to help in making effective irrigation decisions with optimum water usage in smart farming.

Advantages : In this paper the implemented smart farming system was found to be feasible and cost effective for optimizing water resources for agriculture of precision.

Disadvantages: The major drawback is that the need to measure the hardware performance at the server level when ingesting data. The limitations of AI techniques especially in training speed and accuracy balance that may hamper the integration of ML.

Journal paper 3:

Title : Smart farming becomes even smarter with deep learning

Author : Zeynep Ünal

Inference: This paper explores the advantages of using deep learning in agricultural applications. The bibliography reviews the potential of using deep learning techniques in agricultural industries.

Advantages: This paper identified where deep learning has been used for improving various agricultural practices.

Disadvantages: In this paper the UAV aided studies were not analyzed as an independent topic and UAV can be considered as an integral part of smart farming.

Journal paper 4:

Title: Machine Learning for Smart Farming: A Focus on Desert Agriculture

Author: Anandhavalli Muniasamy

Inference: This paper addresses the supports provided by IoT and machine learning techniques in agriculture data collection, processing, and decision-making services for smart farming.

Advantages: In this paper, in order to explore the the scope of machine learning and smart farming in desert agriculture, farm management information systems are used.

Disadvantages : Big data due to the growing flow of data from sensors for the integration of satellite, drone, seed drilling, field map, etc., which requires greater processing power. Lack of standards for data integration, ownership and interoperability of various services.

Journal paper 5:

Title: Solar energy harvesting for smart farming using nanomaterial and machine learning

Author: Rambabu Vatti, Nagarjuna Vatti , K Mahender, Prasanna Lakshmi Vatti and B. Krishnaveni

Inference: This paper presents a more efficient solar energy harvesting technology which uses nanomaterial for improving conversion efficiency and machine learning technology to maximize the collection of solar radiation by continuously tracking the path of the SUN in all the seasons.

Advantages: This paper focuses on increasing the conversion efficiency of the solar energy systems using nanomaterial and the energy collection efficiency is increased by training the smart farm controllers using machine learning algorithms.

Disadvantages: The main draw backs of the solar cells made up of silicon is low conversion efficiency and bigger in size so is the cost.

2.4 Software requirements:

Hardware Requirements:

Processor: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz

Installed RAM: 8.00 GB (7.74 GB usable)

System type: 64-bit operating system, x64-based processor 3.3.2

Software Requirements:

Anaconda Jupyter notebook

Operating systems windows 10

CHAPTER 3

PROJECT DESIGN

3.1 Description about the project:

A comparative study of various machine learning can be applied on a dataset with a view to determine the best performing methodology. The prediction is found by applying the Regression Based Techniques such as Logistic, Random Forest and support vector machine. These machine learning algorithms are performed on the dataset containing details about the climatic conditions, Temperature, humidity present in the atmosphere and the ph levels.

In this project, the model is given with the data from the environment such as the temperature, humidity present in the atmosphere and the ph values as the input and predicts the crop that is best suited for that particular weather conditions. To implement the project, a dataset is taken. To create the dataset, information about crops over the previous years was gathered from a variety of sources, such as government websites.

Firstly, the dataset is cleaned by removing the null values and treating the missing values. After data cleaning, data preprocessing takes place which is the most important part. After data preprocessing, machine learning algorithms like Logistic regression, Support vector machine and Random forest are applied to the training data and the model is trained.

During the training process, having a sufficient amount of data is important for the success of the network. With the development of the Internet, it has become easy to provide the amount of data required for training the machine learning models.

Large amounts of data provide an opportunity of developing many approaches to improve the learning performance of the machine learning algorithms. After the model is trained, it is evaluated with the test dataset. The algorithm with more accuracy is said to be accurate in estimating the crop.

Crop yield prediction using machine learning can revolutionize agriculture by enabling data-driven decisions and sustainable farming practices. By leveraging historical data and advanced algorithms, we can enhance crop productivity and address food security challenges.

3.2 Architecture diagram:

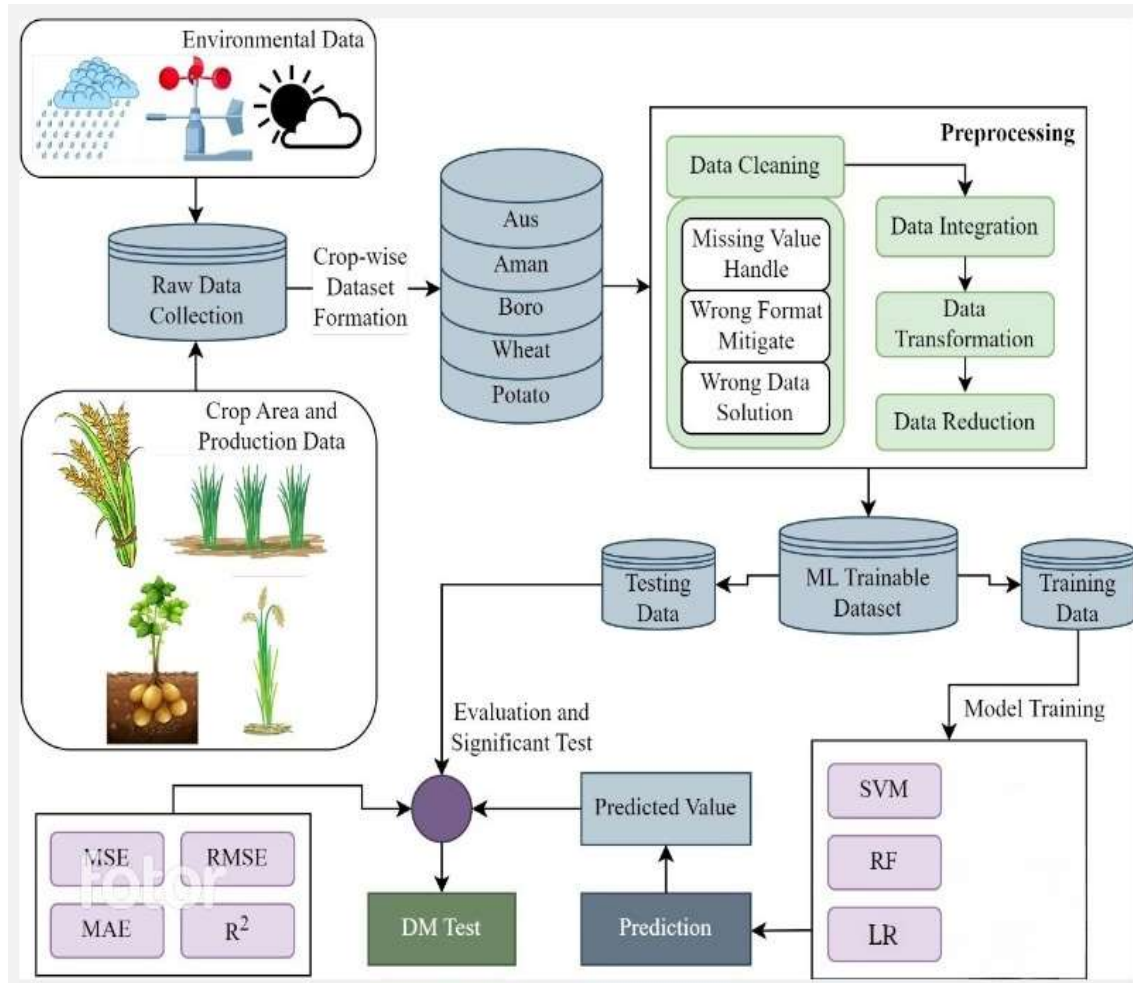


Figure 3.1 - Architecture Diagram of Crop Prediction

Environmental Variables:

Climate variables such as temperature, humidity, wind velocity, rainfall, sunlight hours, water levels, soil content, and soil type significantly impact crop yield.

Data Availability:

Availability of historical data on crop yields, weather conditions, and soil quality is essential for training accurate models.

Feature Engineering:

Identifying relevant features (such as climate data, soil properties, and crop types) and transforming them into meaningful representations for ML models.

Model Selection:

Choosing the right ML algorithm (e.g., Random Forest, Support Vector Machine, or Neural Networks) based on the problem and available data.

Data Collection and Cleaning:

- Gather historical data on crop yields, climate conditions, and soil properties.
- Clean the data by handling missing values, outliers, and inconsistencies.

Feature Extraction and Preprocessing:

- Extract relevant features (temperature, humidity, pH levels, etc.) from the dataset.
- Normalize or standardize the features to ensure consistent scaling.

Model Training and Evaluation:

- Split the dataset into training and testing subsets.
- Train each regression model using the training data.
- Evaluate model performance using metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared (R^2).

Crop Recommendation:

- Given environmental inputs (temperature, humidity, pH), predict the best-suited crop using the trained model.
- Consider crop-specific requirements (e.g., rice prefers high humidity, wheat prefers moderate temperatures).

Model Comparison:

- Compare the performance of Logistic Regression, Random Forest, and Support Vector Machine.
- Choose the model with the highest accuracy for crop yield prediction.

3.3 Block diagram:

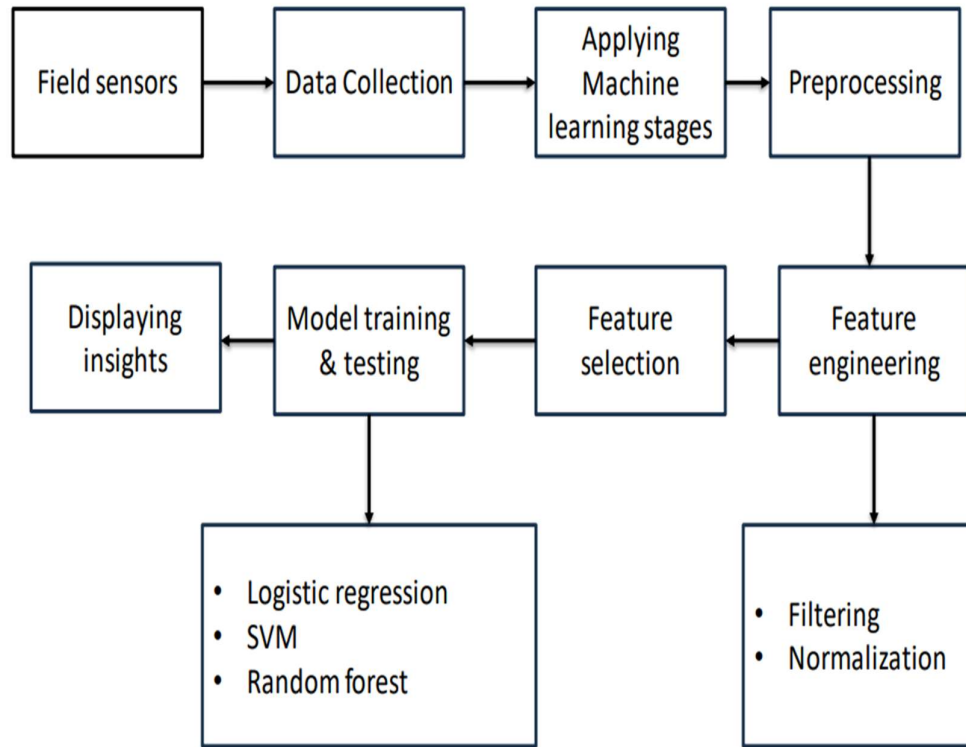


Figure 3.2 – Block Diagram of crop prediction

A comparative study of various machine learning can be applied on a dataset with a view to determine the best performing methodology. The prediction is found by applying the Regression Based Techniques such as Logistic, Random Forest and support vector machine. These machine learning algorithms are performed on the dataset containing details about the climatic conditions, Temperature, humidity present in the atmosphere and the ph levels.

CHAPTER 4

IMPLEMENTATION

4.1 Data collection module:

It includes data collection and understanding the data to study the hidden patterns and trends which helps to predict and evaluating the results. Dataset carries 1401 rows i.e., total number of data and 6 columns i.e., total number of features. Features include temperature, humidity, ph value, water available, season, label of the crop .

4.2 Data preprocessing:

Data preprocessing is most important process. Mostly farming related data contains missing values and other impurities that can cause effectiveness of data. To improve quality and effectiveness obtained after mining process, Data preprocessing is done. To use Machine Learning Techniques on the dataset effectively this process is essential for accurate result and successful prediction. Data preprocessing is done in two steps.

4.2.1 Missing values removal:

Remove all the instances that have zero (0) as worth. Having zero as temperature or humidity or ph is not possible. Therefore this instance is eliminated. Through eliminating irrelevant features/instances we make feature subset and this process is called features subset selection, which reduces dimensionality of data and help to work faster.

4.2.2 Splitting of data:

After cleaning the data, data is normalized in training and testing the model. When data is splitted then we train algorithm on the training data set and keep test data set aside. This training process will produce the training model based on logic and algorithms and values of the feature in training data. Basically aim of normalization is to bring all the attributes under same scale.

4.3 Feature engineering and selection:

Feature engineering is a crucial step in the process of building machine learning models. It involves selecting, transforming, and creating new features from raw data to improve the performance of a model in making predictions or solving a specific task. Feature engineering requires creativity to extract relevant information and represent it in a way that enhances the model's ability to learn patterns and make accurate predictions. Feature selection identifies the most relevant features that contribute the most to the predictive power of the model.

Filtering is a process of selecting or excluding features based on certain criteria without involving the model training process. It is performed as a preprocessing step to reduce the dimensionality of the feature space, remove noise, or improve model interpretability and generalization.

Normalization is a fundamental step in feature engineering that involves scaling the values of features to a similar range. This process ensures that all features contribute equally to the learning process of machine learning algorithms, preventing features with larger scales from dominating the learning process. Normalization does not change the shape of the distribution of the data but rather rescales it.

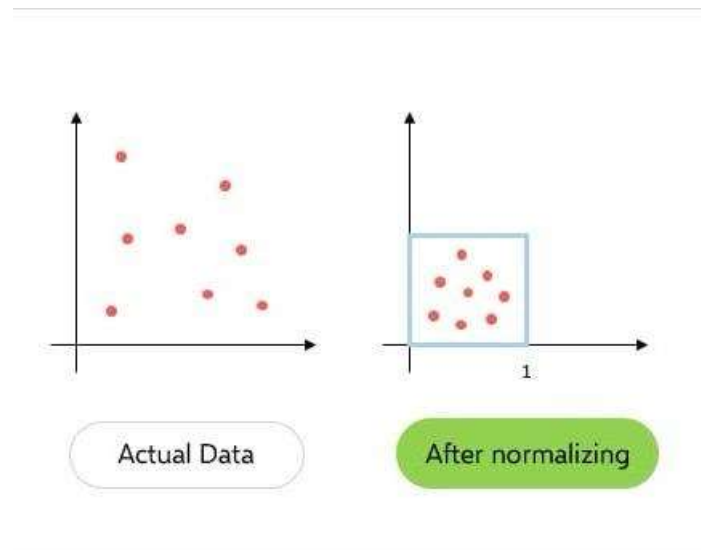


Figure 4.1 – Normalization of data

4.4 Model training and testing:

To train the model we used various machine learning algorithms like Logistic regression, support vector machine and Random forest.

Logistic Regression:

Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lies between 0 and 1.

Support vector machine (SVM):

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. SVM intends to classify the data points by maximizing the margin among classes in a high-dimensional space.

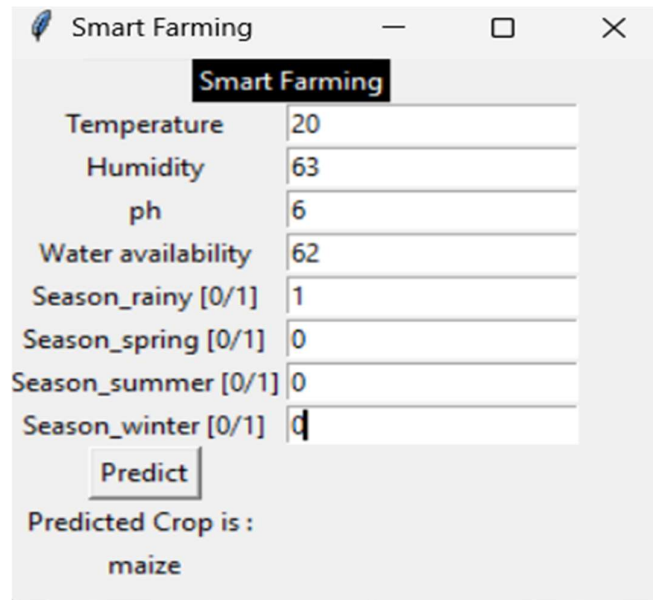
Random forest:

Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique. It can be used for both Classification and Regression problems in ML. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and prove the performance of the model. As the name suggests, “Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset.”

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Outputs:



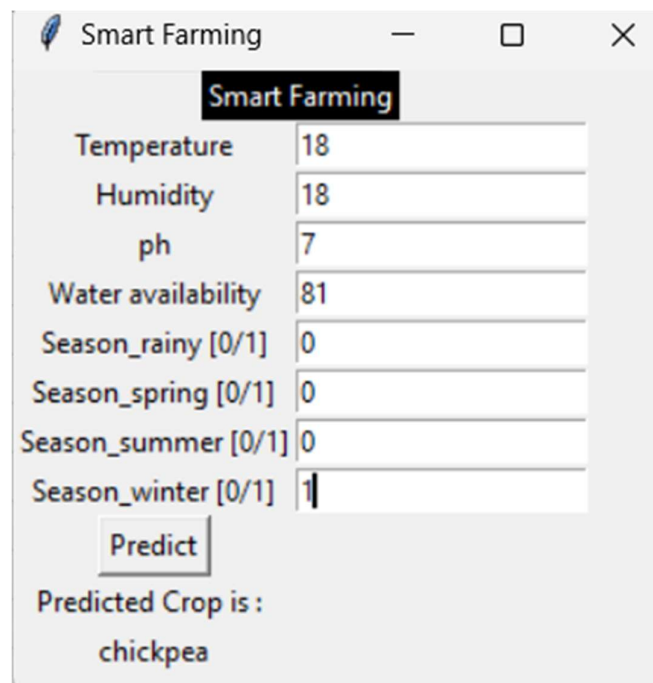
Smart Farming

Smart Farming	
Temperature	20
Humidity	63
ph	6
Water availability	62
Season_rainy [0/1]	1
Season_spring [0/1]	0
Season_summer [0/1]	0
Season_winter [0/1]	0

Predict

Predicted Crop is :
maize

Figure 5.1- Maize crop is predicted when the conditions are given as above



Smart Farming

Smart Farming	
Temperature	18
Humidity	18
ph	7
Water availability	81
Season_rainy [0/1]	0
Season_spring [0/1]	0
Season_summer [0/1]	0
Season_winter [0/1]	1

Predict

Predicted Crop is :
chickpea

Figure 5.2 – Chickpea crop is predicted when the conditions are given as above

Smart Farming

Temperature	22
Humidity	21
ph	5
Water availability	122
Season_rainy [0/1]	0
Season_spring [0/1]	1
Season_summer [0/1]	0
Season_winter [0/1]	0

Predict

Predicted Crop is :
kidneybeans

Figure 5.3 – Kidney beans crop is predicted when the conditions are given as above

Smart Farming

Temperature	37
Humidity	27
ph	5
Water availability	158
Season_rainy [0/1]	1
Season_spring [0/1]	0
Season_summer [0/1]	0
Season_winter [0/1]	0

Predict

Predicted Crop is :
pigeonpeas

Figure 5.4 – Pigeonpeas crop is predicted when the conditions are given as above

Smart Farming

Smart Farming

Temperature: 28

Humidity: 92

ph: 6

Water availability: 24

Season_rainy [0/1]: 0

Season_spring [0/1]: 0

Season_summer [0/1]: 1

Season_winter [0/1]: 0

Predict

Predicted Crop is : muskmelon

Figure 5.5 -Muskmelon crop is predicted when the conditions are given as above

5.2 Comparison of accuracy between ML algorithms:

ML Algorithm	Accuracy
Logistic Regression	88.8%
Support vector machine	98.2%
Random forest	98.5%

Table 5.1 – Accuracy of algorithms

After applying the machine learning algorithms to the model, the accuracies of specific algorithms are as above. Logistic regression is 88.8% accurate, Support vector machine (SVM) is 98.2% accurate and Random forest is 98.5% accurate. The Random forest algorithm is more accurate in predicting the crop with the accuracy rate of 98.5% compared to the other two machine learning algorithms.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion:

The project's findings on crop yield prediction, disease detection pave the way for a future where farms are more data-driven, efficient, and sustainable. However, there's still significant room for future work. By exploring areas like multimodal learning, explainable AI, and end-to-end automation, we can further empower farmers with intelligent decision-making tools. Overall, this project represents a significant step towards a future where smart farming powered by machine learning plays a crucial role in ensuring food security for a growing global population.

6.2 Future work:

Develop machine learning models that can explain their reasoning to farmers. This builds trust and allows farmers to adapt the AI's recommendations to their specific context. Move beyond individual tasks like weed detection. Train AI using reinforcement learning to perform multi-step actions like automatically deciding when and how much pesticide to apply. Implementation of cloud techniques for online usage purpose.

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